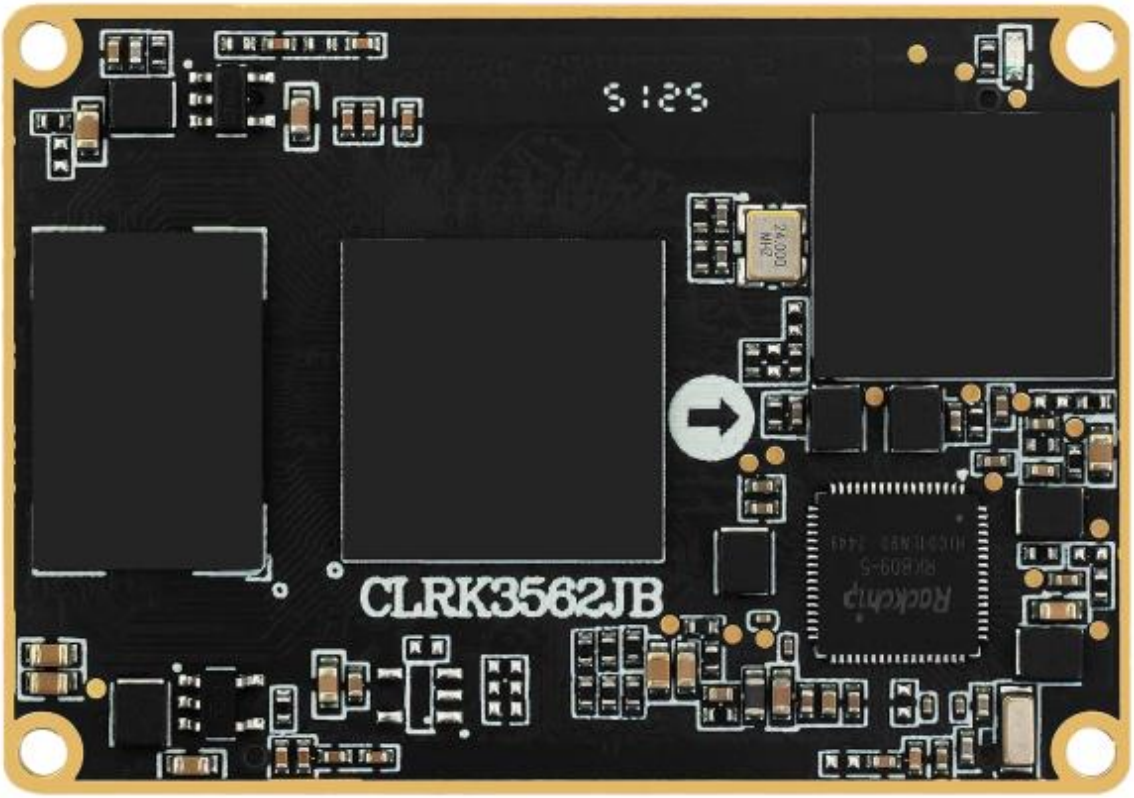
ATK-CLRK3562J

Core Board Specification

V1.0



ALIENTEK

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In order to get the latest version of product information, please regularly visit the download center or contact the customer service of Taobao ALIENTEK flagship store. Thank you for your tolerance and support.

Revision History:

Version	Version Update Notes	Responsible person	Proofreading	Date
V1.0	release officially	ALIENTEK Linux Team	ALIENTEK Linux Team	2026.03.20

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Chapter 1. Core board overview

1.1 Core board introduction

The ALIENTEK ATK-CLRK3562JB Core Board is an embedded industrial-grade core board built on Rockchip RK3562J high-performance, low-power processor.

It adopts a multi-core heterogeneous architecture, featuring a quad-core ARM Cortex-A53 CPU with a main frequency up to 1.8 GHz and a single-core Cortex-M0 MCU running at 200 MHz.

Integrated with a 1.0 TOPS NPU, Mali-G52-2EE GPU and a 13M ISP, the core board supports operating systems such as Linux and Android to meet the demands of complex application scenarios.

Furthermore, it is equipped with abundant peripheral interfaces, including UART, I2C, I2S, SPI, CAN, SDMMC, LVDS / MIPI DSI, MIPI CSI, USB 2.0, USB 3.0 / PCIE 2.1, ADC, 100Mbps Ethernet and Gigabit Ethernet.

It is widely applied in multiple fields: industrial human-machine interfaces, data acquisition terminals, IoT edge computing, smart home systems, smart white goods, building automation control, smart commercial displays, electronic lockers, new energy industries, power control, industrial control, industrial gateways and more.

Comprehensive development resources are provided for the core board.

Software resources include but are not limited to U-Boot source code, Linux kernel source code, peripheral driver source code, file system source code, QT interface source code, Linux application source code, as well as relevant development tools and environments.

Document resources cover dozens of detailed technical manuals and auxiliary documents, such as SDK development handbooks, embedded Linux driver development guides, Linux application hardware design examples, and embedded AI development manuals. These materials are easy to understand and highly practical.

Hardware resources are fully complete, including core board specifications, hardware reference manuals, minimal system hardware design references, baseboard schematic diagrams and original projects, core board interface datasheets, baseboard component footprint libraries, mechanical dimension drawings, chip reference manuals, etc., fully satisfying developers' requirements.

ALIENTEK core boards come with detailed, abundant and highly practical technical resources, helping customers shorten the development cycle and accelerate product launch to the market.

1.2 Typical Application Fields

Widely applicable to industrial human-machine interfaces, data acquisition terminals, IoT edge computing, smart homes, smart white goods, building control, smart commercial displays, electronic storage lockers, new energy, power control, industrial control, industrial gateways and other fields.

Chapter 2. Core Board Resources

2.1 Hardware Resources

The ATK-CLRK3562JB core board is based on the high-performance, low-power Rockchip RK3562J processor.

It is onboard with LPDDR4/4X memory and eMMC storage, delivering efficient data processing capability and supporting large-capacity high-speed storage for embedded systems.

Meanwhile, the core board is equipped with a high-efficiency dedicated PMIC power management chip to meet the stable power supply requirements of high-performance systems.

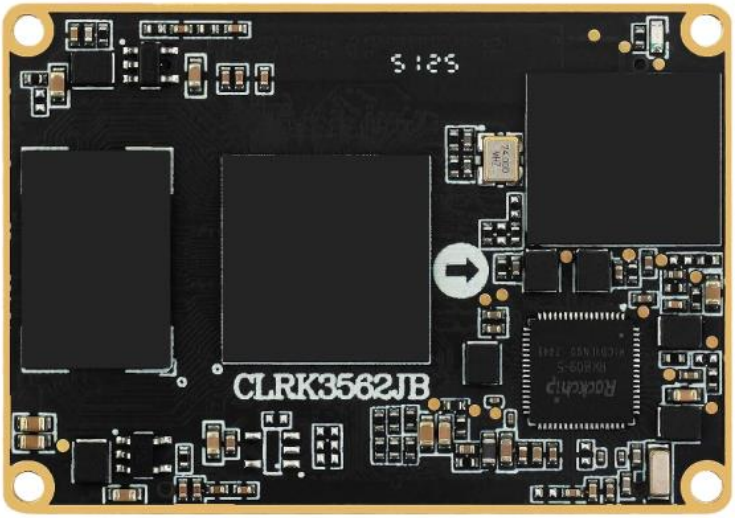


Figure 2.1-1 Front side of the core board

Figure 2.1-2 Back side of the core board

2.2 Processor Resources

The block diagram of RK3562J processor specifications is shown in the figure below.

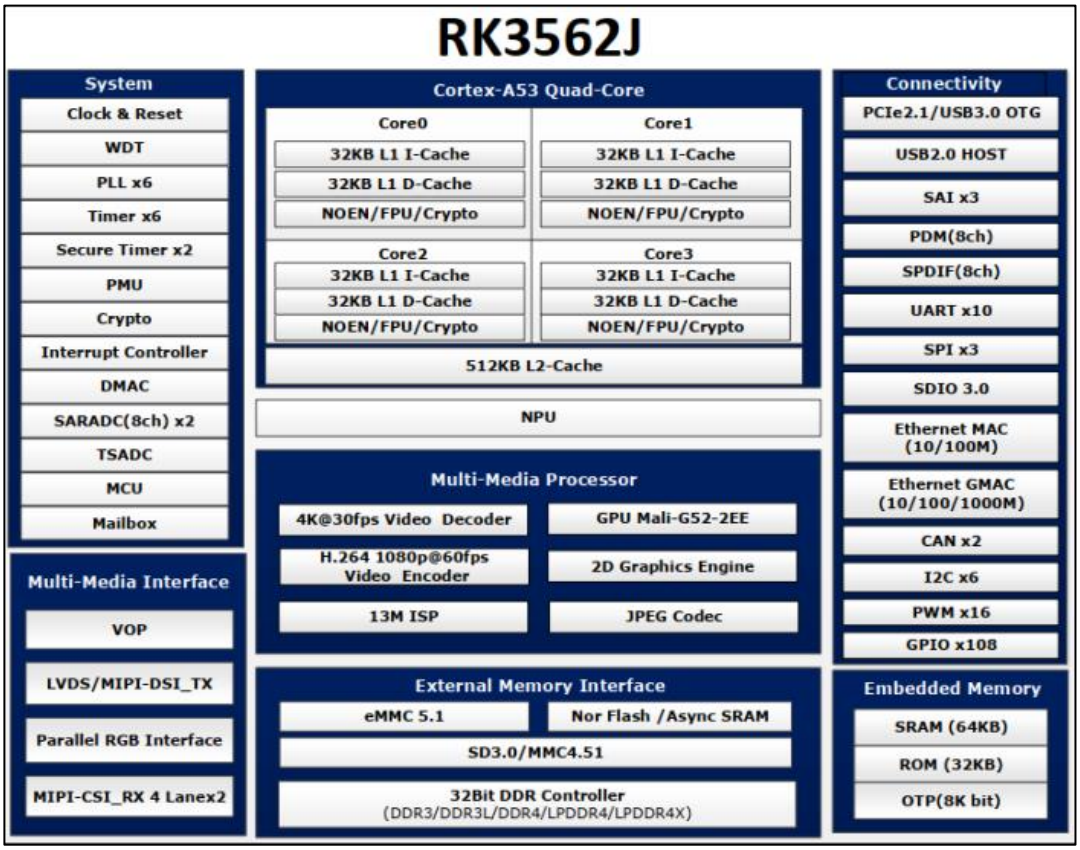


Figure 2.2-1 RK3562J Processor Block Diagram

The main internal resources of the RK3562J processor are listed in the table below. For more detailed specifications, please refer to the RK3562J processor datasheet.

Main Resources of RK3562J Processor			
CPU	Quad-core Cortex-A53, 1.8GHz Single-core Cortex-M0, 200MHz	**GPU**	Mail-G52-2EE
NPU	1.0 TOPS	**ISP**	13M, maximum resolution 4224×3136 Supports 3A, HDR, 3DLUT, etc.
3D Graphics Engine	OpenGL ES 1.1/2.0/3.2、OpenCL 2.0、Vulkan 1.0/1.1	**2D Graphics Engine**	Format conversion, image scaling, image rotation, alpha blending, etc.
Video Encoder	H.264 1920x1080@60fps	**Video Decoder**	H.264 1920x1080@60fps\H.265/VP9 4096x2304@30fps
JPEG Encoder	Max resolution up to 8192×8192 Encoding compression: 90 megapixels per second	**JPEG Decoder**	Up to 65536 × 65536 Supports standard decoding, DRI decoding, etc.

MIPI CSI RX	2 lanes with 4 data channels Configurable as 4 lanes with 2 data channels Maximum 2.5 Gbps per lane	**VICAP Video input**	RAW8/10/12/14, YUV422 Supports up to 4 MIPI CSI camera inputs
RGB	2048×1080@60Hz Supports RGB format, up to 8-bit Data rate up to 150 MHz	**BT.656/BT.1120**	BT656 576 P/I output BT.1120 1080 P/I output
MIPI DSI TX	1-channel DSI with 4 data lanes 2048×1080@60Hz Max 1.2 Gbps/lane Supports RGB format	**LVDS**	800x1280@60Hz Supports RGB888/RGB666 Supports VESA/JEIDA data formats
AUDIO	Up to 192 KHz Supports I2S, PCM, TDM protocols, as well as SPDIF, PDM and Audio DSM.	**SDIO**	×3 Supports 1/4/8 bit Supports SDIO 3.0, eMMC 5.1 Up to HS200 / HS400
I2C	×6 Supports 100 Kbit/s, 400 Kbit/s	**SPI**	×3 Supports software-configurable master-slave mode.
UART	×10 5/6/7/8 bits, baud rate up to 4 Mbps	**PWM**	×16 Supports capture mode, continuous mode and single-shot mode.
USB2.0 HOST	×1 High-speed (480 Mbps), Full-speed (12 Mbps), Low-speed (1.5 Mbps)	**CAN**	×2 Supports standard frames, extended frames, and 8-byte data frames. Up to 1 Mbps.
USB3.0	×1 (Multiplexed with PCIe) Supports USB 3.0 dual-role master/slave mode switching	**PCIE 2.1**	×1 (Multiplexed with USB 3.0) PCIe 2.1 Supports RC mode only 2.5 Gbps / 5 Gbps per lane supported
Fast Ethernet RMII	×1 10/100 Mbps speed Supports full-duplex and half-duplex modes	**Gigabit Ethernet RGMII**	×1 10/100/1000 Mbps speed Supports full-duplex and half-duplex modes
GPIO	×108 (Multiplexed with other functions) Supports interrupt trigger, pull-up/pull-down	**ADC**	×16 10-bit accuracy Sampling rate up to 1 MS/s

	configuration, and drive strength adjustment.		

Note: These are the resource parameters of the processor in the datasheet, not the available resource parameters of the core board.

The core board exports most of the I/O pins of the processor. Users can design their own baseboard according to their requirements to utilize the I/O resources on the core board and configure the I/O multiplexing functions as needed.

2.3 Hardware Specifications

The hardware specifications of the core board are shown in the table below.

Item	Parameter	Remarks
Core Board Model	ATK-CLRK3562JB	
Dimensions	50mm*35mm	Length × Width
Processor Model	RK3562J	
Processor Architecture	Quad-core Cortex-A53 with a main frequency up to 1.8 GHz, plus a single-core Cortex-M0 running at 200 MHz. It integrates a 1.0 TOPS NPU, GPU and 13M ISP.	
Power Management	Dedicated high-efficiency integrated power management IC	
Memory	LPDDR4/4X: 1GB/2GB optional, customizable in bulk upon request	The actual mounted component shall prevail for the model.
Storage	eMMC: 8GB/32GB optional，supports bulk customization and replacement	The actual mounted component shall prevail for the model.
Operating Power Supply	5V/2A、 3.3V/2A	Recommended power supply tolerance: within 200mV, and shall not exceed 5.5V.
Power Consumption	> 0.5W	Power consumption of the core board in minimal system configuration; the actual power consumption depends on the number of peripherals in use.
Operating Temperature	-40℃ ~ +85℃	
Interface Type	Board-to-board connection, equipped with 2 sets of 120-pin anti-reverse insertion male BTB connectors.	
Interface Pins	Connector pin pitch: 0.5mm, total number of pins: 240PIN	
Core Board Height	The combined height of the connector is 3 mm. After assembly, the overall	

	height of the core board is approximately 6 mm, measured from the highest position of the core board components.	
Reserved Mounting Holes	Four M2 mounting holes	
PCB Process	8-layer PCB, 1.6mm board thickness, immersion gold finish, with separate ground layer and power layer.	

2.4 Software Specifications

The core board supports running various operating systems and multiple drivers, as shown in the table below.

System & Driver Support		
Operating System	Linux Buildroot(Factory Default OS, Recommended)	
	Android 14	
	Debian(Built-in version in SDK, not maintained)	
	RT-Thread, The Cortex-M0 core is operable (SDK built-in version, not maintained)	
Graphical Interface	QT 5.15	
Driver Support	PMIC	EMMC
	LPDDR4/4X	ISP
	NPU	GPU
	LED	RTC
	KEY	I2C
	CAN	SPI
	I2S	RS485 UART
	RS232 UART	USB HOST
	USB OTG	PCIE 2.0
	USB 3.0	PCIE TO NVMe / Ethernet
	AUDIO OUT	MIC IN
	PWM	WATCHDOG
	4G	5G RedCap
	WIFI	BLUETOOTH
	MIPI-CSI camera	Touch Screen
	MIPI-DSI display screen	LVDS display screen
	Fast Ethernet	Gigabit Ethernet
Note	A complete system driver support package and SDK are provided for ease of use and secondary development.	

Chapter 3. Electrical Characteristics

The electrical characteristics parameters of the core board are shown in the table below.

Operating Parameters	Minimum Value	Typical Value	Maximum Value
Core Board Operating Voltage1	/	3.3V	/
Core Board Operating Voltage2	/	5.0	5.5V
Core Board Operating Temperature (Industrial Grade)	-40℃	/	+85℃

The reference power consumption values of the core board are shown in the table below.

No.	Test Case	Input Power of Test Board	Typical Current	Typical Power Consumption
1	Core board powered on, no system running.	12V	37mA	0.44W
2	The core board only runs the factory system without connecting any external modules.	12V	45mA	0.55W

Notes:

1. The above reference power consumption values are measured on the ATK-DLT527B development board with a DC input power supply of 12V/2.5A.

2. The core board is equipped with multiple interfaces. Actual power consumption is related to the number of peripherals used, external modules, and other factors, and should be measured according to actual application scenarios. The data provided here is for reference only.

Chapter 4. Mechanical Structure

4.1 PCB Parameters

Parameters	Core board
PCB Size	50mm * 35mm
PCB Thickness	1.6mm
PCB Layers	8 layers
PCB Technology	Gold plating
Number of Mounting Holes	4 pieces

4.2 Core Board Mechanical Dimensions

The dimensions of the ALIENTEK ATK-CLRK3562JB core board are 50mm*35mm. The mechanical drawing is shown in the figure below.

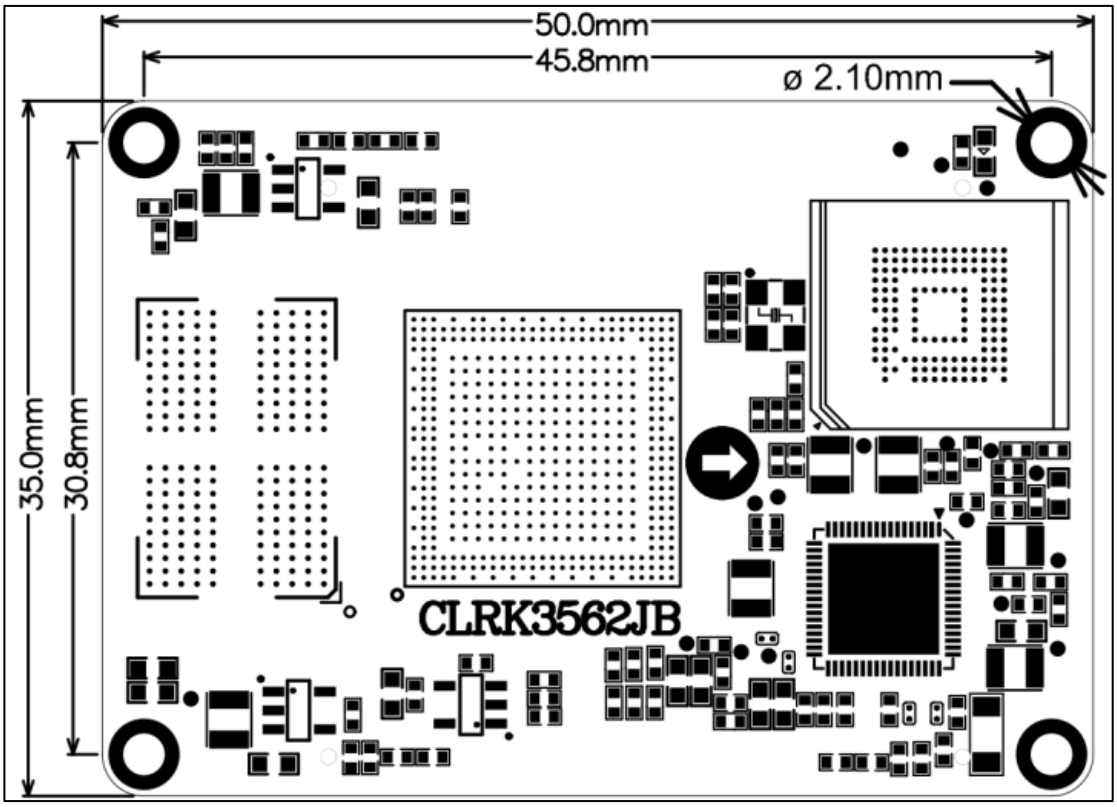


Figure 4.2-1 Core Board Mechanical Drawing

The core board is connected to the base board via B2B (board-to-board) connectors, with a mating height of 3 mm for the connectors.

After all components are soldered and mounted on the core board PCB, the maximum overall height of the PCBA is approximately 6 mm (measured at the tallest component).

Chapter 5. Core Board Resources

ALIENTEK provides abundant development resources for the ATK-CLRK3562JB core board.

Software resources include but are not limited to U-Boot source code, Linux source code, peripheral driver source code, file system source code, QT interface source code, Linux application source code, as well as relevant development tools and development environments.

Document resources cover detailed development documents and auxiliary materials, including dozens of manuals such as the SDK development manual, embedded Linux driver development guide, Linux application hardware development examples, and embedded AI development manual. These documents are easy to understand and highly practical.

Hardware resources are comprehensive, including the core board specification, hardware reference manual, minimal system hardware design reference, baseboard schematic diagrams and original projects, core board interface datasheets, baseboard component footprint libraries, mechanical dimension drawings, chip reference manuals, and more. They fully meet the usage requirements of developers.

All supporting resources of the core board are available for download via cloud disk.

5.1 Resource Download

ATK-CLRK3562JB Core Board Resource Download Center:
http://www.openedv.com/docs/boards/arm-linux/RK3562Jhxb%20Linux.html

5.2 Resource Description

The level-1 directory of the development board resources on NetDisk Drive A in Chinese version is shown in the figure below.

Figure 5.2-1 Level-1 Directory of Development Board Drive A Resources

Figure 5.2-2 Level-1 Directory of Development Board Drive B Resources

For English resources, please refer to the corresponding development board repository on GitHub: https://github.com/openedv/development_board_ATK-DLRK3562J.git

Directory	Description
01_codes	AI routines, heterogeneous-core routines, Linux driver routines, QT graphical interfaces, application hardware development examples, etc.
02_sch	Development Board Base Board Schematic
03_ hardware	Hardware-related materials collection, including baseboard onboard chip documents, development board specifications, development board package libraries, development board mechanical dimensions, certification documents, etc.
04_softwares	Collection of required software tools, including serial terminal software, USB-to-serial driver, etc.
05_tools	Collection of development tools, including cross-compiler, ADB debugging tool, programming tool, Allwinner official development tools, etc.
06_reference_data	Reference materials collection, including protocol manuals, ARM manuals, and other reference documents.
07_RK_ reference	Rockchip Official SDK Documentation
08_system_image	Factory Default System Image Package
09_user_manual	Detailed and comprehensive supporting development documents to facilitate users' rapid development.

Directory Description for Development Board Materials (Disk B):

Disk B Directory	Description
01_Ubuntu Development Environment	Ubuntu virtual machine, applicable for development board SDK source code development and related work.
02_ATK-DLRK3562JB_board_SDK	Factory System SDK Source Code Package for Development Board

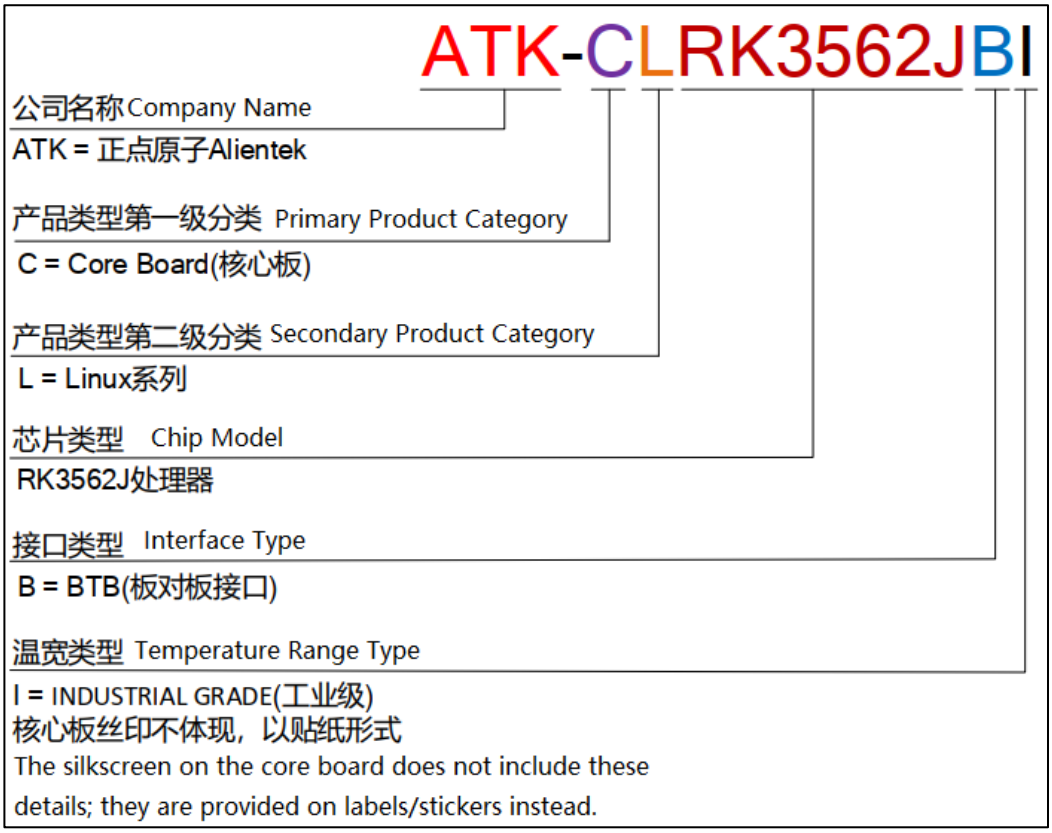
Note: Due to space limitations, the corresponding directory materials are not listed here. Please download the development board documents for detailed reference.

Chapter 6. Core Board Selection

6.1 Model Naming Convention

Core Board Model: ATK-CLRK3562JB.

The model parameter analysis is as follows.



Chapter 7. After sales service

7.1 Terms of after-sales service

1). After receiving the goods, please open them in front of the express, and sign after acceptance. If you find that the goods are less after signing, take photos in time and contact the seller's customer service to explain the situation within 15 days. If the feedback is lack of goods after 15 days, we will not reissue the goods. Other reasons notwithstanding).

2). 15 days -1 month: we are responsible for the return freight repair of product problems. Human factors damage expensive main chip or LCD screen, touch screen. The buyer needs to pay the cost and one time shipping fee, no maintenance fee.

3). 1-3 months: the problem of the product itself (non-human factors), we are responsible for the delivery of the past freight maintenance. If the main chip is burned out and the LCD screen and touch screen are damaged, the buyer needs to pay the cost, and the maintenance fee is not charged.

4) After 3 months: the buyer shall bear the return freight and the cost of chip, LCD screen and touch screen. No service charge.

7.2 After-sales Support

Technical support:

QQ group: ALIENTEK RK Communication group 3: 1079854944

ALIENTEK RK3562J User Group: 1072910286 (order number required)

Taobao shop: ALIENTEK flagship store

Business Contact: **ALIENTEK Flagship Store**, transfer to Business Customer Service

Tel: 020-38271790 (reception transfers to Linux Technical Support)

Forum: http://www.openedv.com/forum-277-1.html